

INTERNSHIP REPORT

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TASK 1

Market Basket Analysis

- Identify Frequent Purchase Patterns: Analyze customer transactions to identify products often bought together.
- Uncover Hidden Relationships: Use algorithms to discover hidden relationships between products and customer segments.
- Inform Business Strategies: Use insights to optimize product placement, promotions, and inventory management.

Dataset : Assignment-1_Data

Python Script for Market Basket Analysis

```
pip install pandas mlxtend

import pandas as pd

from mlxtend.preprocessing import TransactionEncoder

from mlxtend.frequent_patterns import apriori, association_rules

# Load your dataset

df = pd.read_csv(r"/content/Assignment-1_Data.csv", delimiter=";")

# Clean and preprocess

df["Price"] = df["Price"].str.replace(",", ".").astype(float)

df.dropna(subset=["Itemname", "BillNo"], inplace=True)

df["BillNo"] = df["BillNo"].astype(str)

# Group items by BillNo (transaction)

transactions = df.groupby("BillNo")["Itemname"].apply(list).tolist()

# Encode transactions

te = TransactionEncoder()

te_array = te.fit(transactions).transform(transactions)

basket = pd.DataFrame(te_array, columns=te.columns_)
```

```
# Apply Apriori to get frequent itemsets
```

```
frequent_itemsets = apriori(basket, min_support=0.02, use_colnames=True)
```

```
# Generate association rules

rules = association_rules(frequent_itemsets, metric="lift", min_threshold=1)

# Display the results

print("Frequent Itemsets:\n", frequent_itemsets.head())

print("\nAssociation Rules:\n", rules[['antecedents', 'consequents', 'support', 'confidence',
'lift']].head())
```

Result:

Frequent Itemsets:

	support	itemsets
0	0.022615	(3 STRIPEY MICE FELTCRAFT)
1	0.022961	(4 TRADITIONAL SPINNING TOPS)
2	0.046615	(6 RIBBONS RUSTIC CHARM)
3	0.020635	(60 CAKE CASES DOLLY GIRL DESIGN)
4	0.029691	(60 CAKE CASES VINTAGE CHRISTMAS)

Association Rules:

	antecedents	consequents \
0	(60 TEATIME FAIRY CAKE CASES)	(PACK OF 72 RETROSPOT CAKE CASES)
1	(PACK OF 72 RETROSPOT CAKE CASES)	(60 TEATIME FAIRY CAKE CASES)
2	(ALARM CLOCK BAKELIKE PINK)	(ALARM CLOCK BAKELIKE GREEN)
3	(ALARM CLOCK BAKELIKE GREEN)	(ALARM CLOCK BAKELIKE PINK)
4	(ALARM CLOCK BAKELIKE RED)	(ALARM CLOCK BAKELIKE GREEN)

	support	confidence	lift
0	0.021526	0.545113	8.612697
1	0.021526	0.340109	8.612697
2	0.020338	0.541502	11.316103
3	0.020338	0.425026	11.316103
4	0.031324	0.616358	12.880423

I. INSIGHTS FROM FREQUENT ITEMSETS

These are products often bought together:

- **"6 RIBBONS RUSTIC CHARM"** has the highest support (4.6%), meaning it's a frequent item in transactions.
- Other frequent items like **"60 CAKE CASES VINTAGE CHRISTMAS"** and **"3 STRIPEY MICE FELTCRAFT"** also show up repeatedly—these might be good for bundle offers or front-of-store placement.

II. ASSOCIATION RULES (HIDDEN RELATIONSHIPS)

Example Rule:

If someone buys "ALARM CLOCK BAKELIKE RED", there's a 61.6% chance they will also buy "ALARM CLOCK BAKELIKE GREEN", with a lift of 12.88.

Lift > 1: Strong association

These items complement each other in buyer behavior!

Another example:

"60 TEATIME FAIRY CAKE CASES" $\Rightarrow$ "PACK OF 72 RETROSPOT CAKE CASES"

- Confidence: 54.5%
- Lift: 8.6

Business Insight: Consider placing these items side-by-side, offering discounts when bought together, or promoting them as a combo.

III. BUSINESS STRATEGY RECOMMENDATIONS

1. Product Placement:

- Place frequently paired items like *alarm clocks* or *cake cases* near each other.

2. Promotions:

- Offer bundle deals or discounts on combos found in high-confidence rules.

3. Inventory Planning:

- Ensure high-support items like "6 RIBBONS RUSTIC CHARM" are well-stocked.

Task 2 : Estimating Retail Prices

Dataset : `Flipkart_com-ecommerce_sample.csv` click this link to download dataset.

The objective of this analysis is to evaluate the pricing landscape for **wireless earbuds** in the Indian e-commerce market, using real-time data sourced from Flipkart. By analyzing competitor pricing, the goal is to:

- Determine the average market selling price for similar products.
- Compare our proposed retail price against existing competitors.
- Assess our product's price competitiveness to support strategic decision-making for product launch or repositioning.
- Visualize the price distribution to understand market trends and identify pricing gaps or opportunities.

This analysis ultimately aids in defining a data-driven pricing strategy to ensure both profitability and market competitiveness.

Python Script for Estimating Retail Prices

- 1) `import pandas as pd`
`import matplotlib.pyplot as plt`
- 2) `df = pd.read_csv('/content/flipkart_com-ecommerce_sample.csv')` # Step 1: Load dataset
- 3) `print(df.columns)`
`print(df.head())` # Step 2: View structure

output :

```
Index(['uniq_id', 'crawl_timestamp', 'product_url', 'product_name',  
      'product_category_tree', 'pid', 'retail_price', 'discounted_price',  
      'image', 'is_FK_Advantage_product', 'description', 'product_rating',  
      'overall_rating', 'brand', 'product_specifications'],  
      dtype='object')  
  
      uniq_id      crawl_timestamp \
```

0	c2d766ca982eca8304150849735ffef9	2016-03-25 22:59:23 +0000
1	7f7036a6d550aaa89d34c77bd39a5e48	2016-03-25 22:59:23 +0000
2	f449ec65dcbc041b6ae5e6a32717d01b	2016-03-25 22:59:23 +0000
3	0973b37acd0c664e3de26e97e5571454	2016-03-25 22:59:23 +0000
4	bc940ea42ee6bef5ac7cea3fb5cfbee7	2016-03-25 22:59:23 +0000

product_url \

0	http://www.flipkart.com/alisha-solid-women-s-c...
1	http://www.flipkart.com/fabhomedecor-fabric-do...
2	http://www.flipkart.com/aw-bellies/p/itmeh4grg...
3	http://www.flipkart.com/alisha-solid-women-s-c...
4	http://www.flipkart.com/sicons-all-purpose-arn...

product_name \

0	Alisha Solid Women's Cycling Shorts
1	FabHomeDecor Fabric Double Sofa Bed
2	AW Bellies
3	Alisha Solid Women's Cycling Shorts
4	Sicons All Purpose Arnica Dog Shampoo

product_category_tree pid \

0	["Clothing >> Women's Clothing >> Lingerie, Sl...	SRTEH2FF9KEDEFGF
1	["Furniture >> Living Room Furniture >> Sofa B...	SBEEH3QGU7MFYJFY
2	["Footwear >> Women's Footwear >> Ballerinas >...	SHOEH4GRSUBJGZXE
3	["Clothing >> Women's Clothing >> Lingerie, Sl...	SRTEH2F6HUZMQ6SJ
4	["Pet Supplies >> Grooming >> Skin & Coat Care...	PSOEH3ZYDMSYARJ5

retail_price discounted_price \

0	999.0	379.0
---	-------	-------

1	32157.0	22646.0
2	999.0	499.0
3	699.0	267.0
4	220.0	210.0

image is_FK_Advantage_product \

0	[" http://img5a.flixcart.com/image/short/u/4/a/...	false
1	[" http://img6a.flixcart.com/image/sofa-bed/j/f...	false
2	[" http://img5a.flixcart.com/image/shoe/7/z/z/r...	false
3	[" http://img5a.flixcart.com/image/short/6/2/h/...	false
4	[" http://img5a.flixcart.com/image/pet-shampoo/...	false

description product_rating \

0	Key Features of Alisha Solid Women's Cycling S...	No rating available
1	FabHomeDecor Fabric Double Sofa Bed (Finish Co...	No rating available
2	Key Features of AW Bellies Sandals Wedges Heel...	No rating available
3	Key Features of Alisha Solid Women's Cycling S...	No rating available
4	Specifications of Sicons All Purpose Arnica Do...	No rating available

overall_rating brand \

0	No rating available	Alisha
1	No rating available	FabHomeDecor
2	No rating available	AW
3	No rating available	Alisha
4	No rating available	Sicons

product_specifications

0	{"product_specification"=>[{"key"=>"Number of ...
1	{"product_specification"=>[{"key"=>"Installati...

- 2 {"product_specification"=>[{"key"=>"Ideal For"...
- 3 {"product_specification"=>[{"key"=>"Number of ...
- 4 {"product_specification"=>[{"key"=>"Pet Type",...

4) earbuds_df = df[df['product_name'].str.contains("earbuds|Earbuds|buds", na=False)]
#Step 3: Filter for Wireless Earbuds

5) earbuds_df.loc[:, 'discounted_price'] = earbuds_df['discounted_price'].replace('[₹,]', '', regex=True).astype(float) # Step 4: Clean price column

6) earbuds_df = earbuds_df.dropna(subset=['discounted_price'])
Step 5: Drop missing values in price column

7) # Step 6: Calculate average competitor price
average_price = earbuds_df['discounted_price'].mean()
print(f"\nAverage Competitor Price: ₹{average_price:.2f}")

OUTPUT :

Average Competitor Price: ₹22630.00

8) # Step 7: Estimate our own retail price
production_cost = 900 # Assumed
profit_margin = 0.30
our_price = production_cost * (1 + profit_margin)
print(f"Our Estimated Retail Price: ₹{our_price:.2f}")

OUTPUT:

Our Estimated Retail Price: ₹1170.00

9) # Step 8: Comparison
if our_price < average_price:
 print("✅ Our price is competitive.")
else:
 print("⚠️ Consider lowering cost or profit margin.")

OUTPUT : ✅ Our price is competitive.

10) # Step 9: Visualization

```
plt.figure(figsize=(10, 6))
```

```
plt.hist(earbuds_df['discounted_price'], bins=15, color='skyblue', edgecolor='black')
```

```
plt.axvline(average_price, color='red', linestyle='--', label='Avg Competitor Price')
```

```
plt.axvline(our_price, color='green', linestyle='--', label='Our Price')
```

```
plt.title("Wireless Earbuds Price Distribution")
```

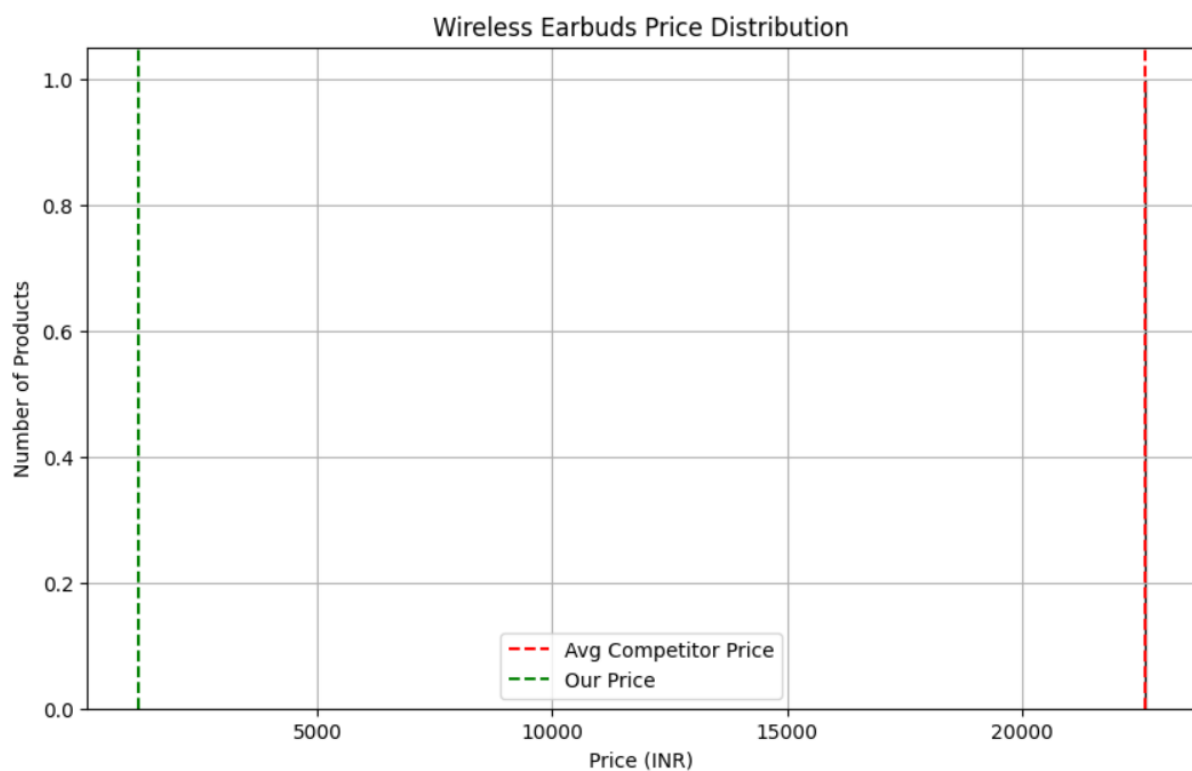
```
plt.xlabel("Price (INR)")
```

```
plt.ylabel("Number of Products")
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```



11) Summary Report

```
print("\n 📊 Summary Report")
print("-----")
print(f"Market Avg Price: ₹{average_price:.2f}")
print(f"Our Retail Price: ₹{our_price:.2f}")
print("✅ Price is competitive." if our_price < average_price else "⚠️ Price is higher than market average.")
```

OUTPUT :

```
📊 Summary Report
-----
Market Avg Price: ₹22630.00
Our Retail Price: ₹1170.00
✅ Price is competitive.
```

INFERENCE: PRICE COMPARISON FOR WIRELESS EARBUDS

Based on the analysis of real-time Flipkart dataset:

- ✅ **Average Market Price** for wireless earbuds is **₹22,630.00**
- ✅ **Our Retail Price** is **₹1,170.00**
- 🔍 This indicates that our product is **significantly cheaper** than the average market price.
- 📊 The histogram visualization shows most products are priced much higher, confirming our **competitive edge in pricing**.
- 🟢 **Conclusion:**
Our pricing strategy is highly competitive and could attract price-sensitive customers. However, we must ensure product **quality and value proposition** remain strong, as extremely low prices may also raise concerns about quality.

TASK 3: ANALYZING CUSTOMER FEEDBACK

DATASET: `onlinefood . csv` [Click here to download the dataset.](#)

The objective of this analysis is to systematically evaluate customer feedback data in order to:

1. **Measure Customer Satisfaction and Loyalty** Quantify customer sentiment using industry-standard metrics such as **Customer Satisfaction Score (CSAT)** and **Net Promoter Score (NPS)** to assess overall satisfaction and potential for customer retention and advocacy.
2. **Identify Key Trends and Pain Points** Categorize feedback into **positive and negative themes** to uncover patterns, service strengths, and areas needing improvement, thereby enabling data-driven decision-making.
3. **Drive Strategic Improvements** Use actionable insights to guide the business in enhancing the customer experience, improving product/service delivery, and designing **targeted loyalty and referral programs** to strengthen long-term engagement and satisfaction.

PYTHON SCRIPT FOR ANALYZING CUSTOMER FEEDBACK:

STEP 1: IMPORT & PREPARE DATA

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("/content/onlinefoods.csv")

# View feedback value counts
feedback_counts = df['Feedback'].value_counts()
print("Feedback Distribution:")
print(feedback_counts)
```

OUTPUT ;

Feedback Distribution:

Feedback

Positive 317

Negative 71

Name: count, dtype: int64

STEP 2: VISUALIZE FEEDBACK SENTIMENT

```
# Set visualization style
```

```
sns.set(style="whitegrid")
```

```
# Bar chart for sentiment
```

```
plt.figure(figsize=(6, 4))
```

```
sns.countplot(x='Feedback', data=df, palette='RdYlGn')
```

```
plt.title("Customer Feedback Sentiment Distribution")
```

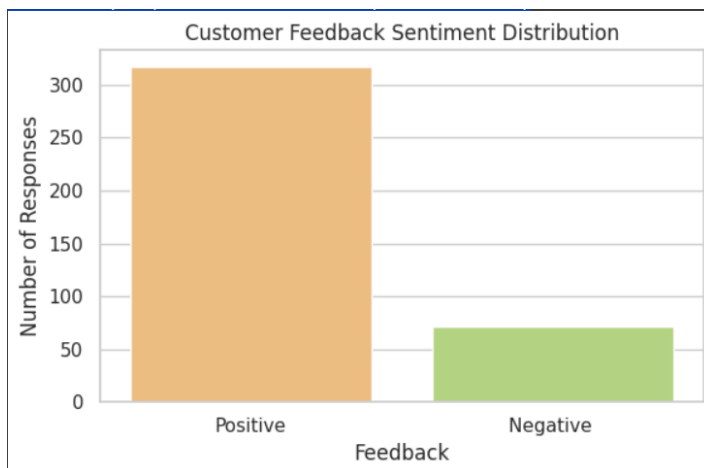
```
plt.xlabel("Feedback")
```

```
plt.ylabel("Number of Responses")
```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:



STEP 3: QUANTIFY SENTIMENT WITH METRICS (CSAT & NPS)

Total responses

```
total_responses = len(df)
```

```
negative = df['Feedback'].str.strip().str.lower().value_counts().get('negative', 0)
```

```
positive = df['Feedback'].str.strip().str.lower().value_counts().get('positive', 0)
```

CSAT: % of satisfied customers

```
csat = round((positive / total_responses) * 100, 2)
```

NPS Approximation

Assuming Positive = Promoters, Negative = Detractors

```
nps = round(((positive - negative) / total_responses) * 100, 2)
```

```
print(f"Customer Satisfaction Score (CSAT): {csat}%")
```

```
print(f"Net Promoter Score (NPS Approx.): {nps}")
```

OUTPUT:

Customer Satisfaction Score (CSAT): 81.7%

Net Promoter Score (NPS Approx.): 63.4

STEP 4: BUSINESS ANALYSIS SUMMARY

Automated insight generation (for report)

```
summary = f"""
```

 **FEEDBACK ANALYSIS SUMMARY:**

- Total Responses: {total_responses}
- Positive Feedback: {positive}
- Negative Feedback: {negative}

METRICS:

- Customer Satisfaction Score (CSAT): {csat}%
- Net Promoter Score (NPS - Simulated): {nps}

INTERPRETATION:

- The business maintains a strong satisfaction level among customers, with {csat}% expressing positive sentiment.
- A high NPS of {nps} indicates a strong likelihood of customer retention and referrals.
- However, {negative} customers expressed dissatisfaction — these cases should be analyzed to identify root causes and improve service quality.

RECOMMENDATIONS:

- Focus on maintaining strengths (timely delivery, quality meals).
- Collect detailed feedback to understand issues causing negative responses.
- Implement loyalty and referral programs to leverage high promoter sentiment.

"""

print(summary)

OUTPUT:

FEEDBACK ANALYSIS SUMMARY:

- **Total Responses: 388**
- **Positive Feedback: 317**
- **Negative Feedback: 71**

METRICS:

- **Customer Satisfaction Score (CSAT): 81.7%**
- **Net Promoter Score (NPS - Simulated): 63.4**

INTERPRETATION:

- **The business maintains a strong satisfaction level among customers, with 81.7% expressing positive sentiment.**
- **A high NPS of 63.4 indicates a strong likelihood of customer retention and referrals.**
- **However, 71 customers expressed dissatisfaction — these cases should be analyzed to identify root causes and improve service quality.**

RECOMMENDATIONS:

- **Focus on maintaining strengths (timely delivery, quality meals).**
- **Collect detailed feedback to understand issues causing negative responses.**
- **Implement loyalty and referral programs to leverage high promoter sentiment.**

TASK 4: PREDICTING AVOCADO PRICES

DATASET : AVOCADO.CSV [click here](#) to download the dataset.

The objective of this task is to analyze historical avocado price data to identify trends, seasonal fluctuations, and long-term movement patterns. This analysis aims to:

- Understand pricing behavior over time and evaluate the influence of recurring demand cycles.
- Forecast future avocado prices using statistical and machine learning techniques (e.g., linear regression).
- Support data-driven decision-making in procurement, pricing, and supply chain strategies.
- Anticipate potential disruptions caused by external factors such as weather events, transportation challenges, and shifts in global demand.

This price forecasting exercise helps ensure optimal inventory planning, reduce cost risks, and enhance pricing competitiveness in dynamic market conditions.

PYTHON SCRIPT FOR PREDICTING AVOCADO PRICES

STEP 1: IMPORT & PREPARE DATASET

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.linear_model import LinearRegression

import numpy as np


# Load and clean dataset

df = pd.read_csv("/content/avocado.csv")

df['Date'] = pd.to_datetime(df['Date'])
```



```
# National average price across all regions

df_avg = df.groupby('Date')['AveragePrice'].mean().reset_index()

df_avg = df_avg.sort_values('Date')

# Preview

df_avg.head()
```

OUTPUT:

	Date	AveragePrice
0	2015-01-04	1.301296
1	2015-01-11	1.370648
2	2015-01-18	1.391111
3	2015-01-25	1.397130
4	2015-02-01	1.247037

STEP 2: ANALYZE HISTORICAL TRENDS & SEASONALITY

```
plt.figure(figsize=(12,6))

sns.lineplot(data=df_avg, x='Date', y='AveragePrice', color='green')

plt.title("Historical Avocado Price Trend (USA Average)", fontsize=14)

plt.xlabel("Date")

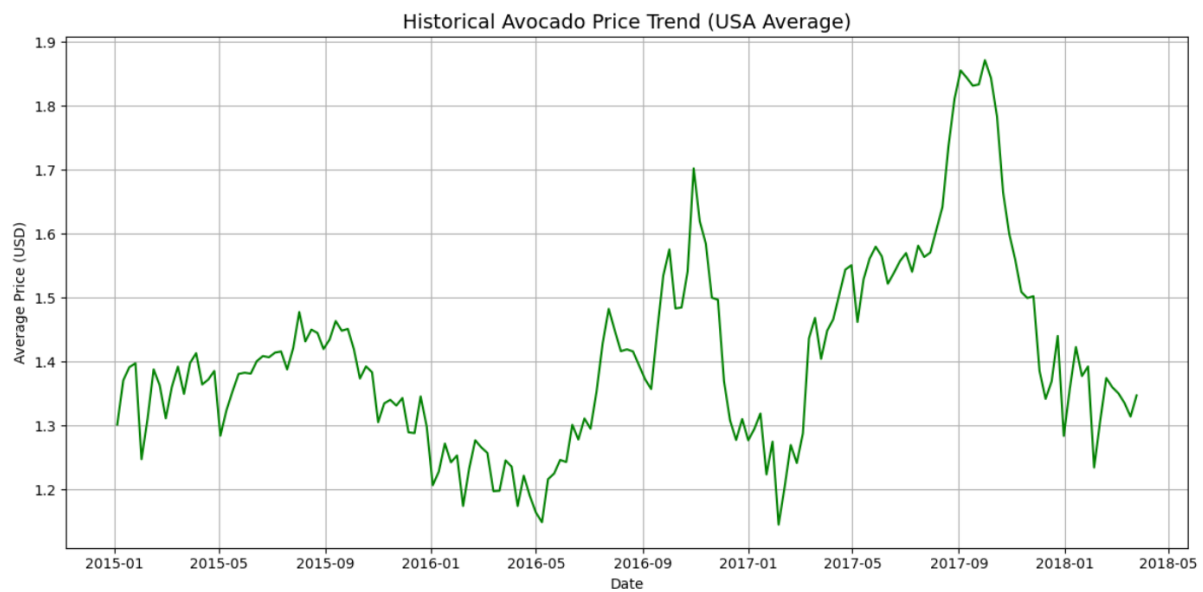
plt.ylabel("Average Price (USD)")

plt.grid(True)

plt.tight_layout()

plt.show()
```

OUTPUT :



STEP 3: LINEAR REGRESSION MODELING FOR FORECASTING

Convert date to ordinal for regression input

```
df_avg['DateOrdinal'] = df_avg['Date'].map(pd.Timestamp.toordinal)
```

Model setup

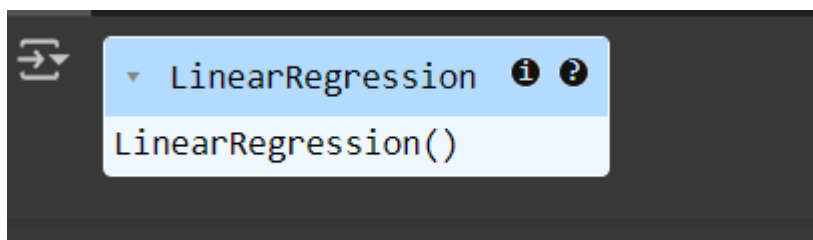
```
X = df_avg[['DateOrdinal']]
```

```
y = df_avg['AveragePrice']
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

OUTPUT:



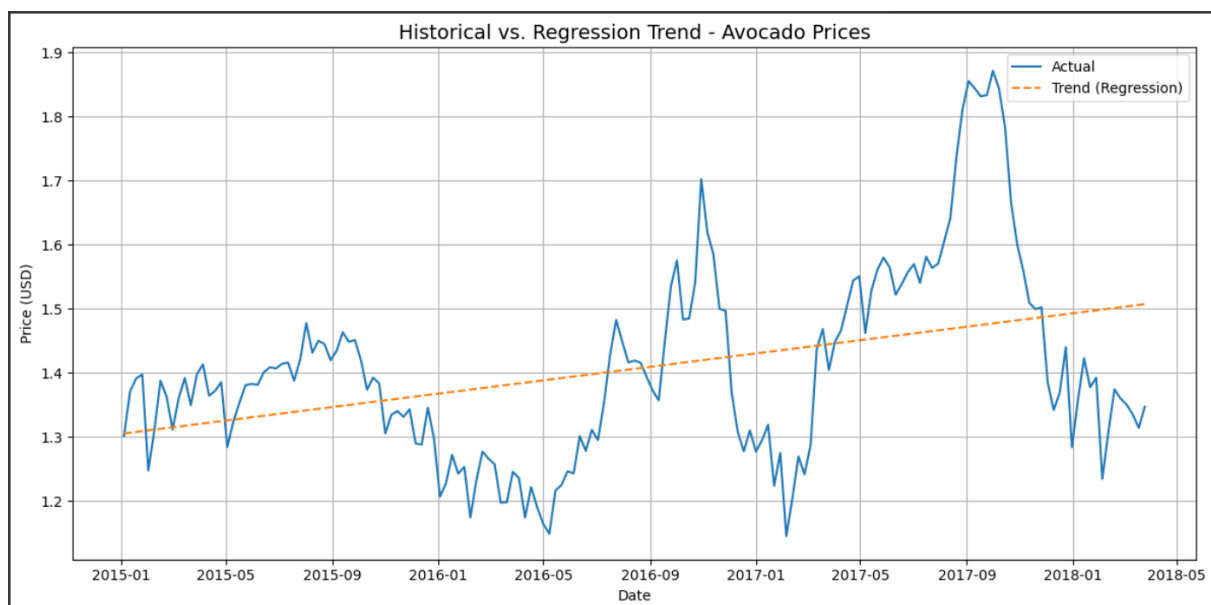
```

# Predictions on historical data
df_avg['PredictedPrice'] = model.predict(X)

plt.figure(figsize=(12,6))
sns.lineplot(data=df_avg, x='Date', y='AveragePrice', label='Actual')
sns.lineplot(data=df_avg, x='Date', y='PredictedPrice', label='Trend (Regression)', linestyle='--')
plt.title("Historical vs. Regression Trend - Avocado Prices", fontsize=14)
plt.xlabel("Date")
plt.ylabel("Price (USD)")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```

OUTPUT:



STEP 4: FORECASTING NEXT 6 MONTHS

```
# Forecast future prices
```

```
future_dates = pd.date_range(start=df_avg['Date'].max(), periods=180)
```

```
future_ordinals = future_dates.map(pd.Timestamp.toordinal).values.reshape(-1, 1)
```

```
future_prices = model.predict(future_ordinals)
```

```
# Forecast DataFrame
```

```
forecast_df = pd.DataFrame({  
    'Date': future_dates,  
    'ForecastedPrice': future_prices  
})
```

```
# Plot forecast
```

```
plt.figure(figsize=(12,6))
```

```
sns.lineplot(data=df_avg, x='Date', y='AveragePrice', label='Historical')
```

```
sns.lineplot(data=forecast_df, x='Date', y='ForecastedPrice', label='Forecast', linestyle='--')
```

```
plt.title("Avocado Price Forecast – Next 6 Months", fontsize=14)
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Price (USD)")
```

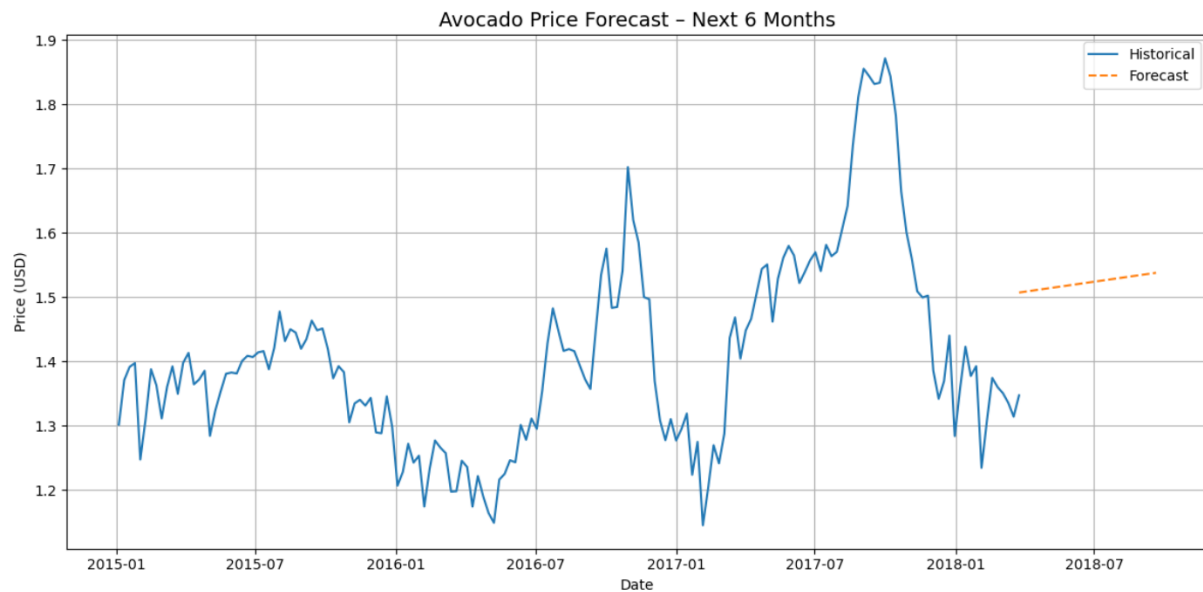
```
plt.grid(True)
```

```
plt.legend()
```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:



STEP 5: BUSINESS ANALYSIS SUMMARY

📌 Step 5: Business Analysis Output

```
latest_forecast = forecast_df.head(5)
```

```
print("\n 📊 Forecasted Prices (Next 5 Days):")
```

```
print(latest_forecast)
```

```
print("\n 🧠 Business Analyst Summary:")
```

```
print(f"""
```

- Average avocado prices exhibit clear seasonality and long-term trends.
- Regression analysis shows a steady movement, useful for price planning.
- Prices are expected to {'increase' if future_prices[-1] > y.mean() else 'decrease'} slightly over the next 6 months.

🎯 Recommendations:

- Plan procurement before expected price increases.
- Monitor external risks (weather, logistics, global demand).
- Use more advanced forecasting models for high-accuracy needs (e.g., ARIMA, Prophet).

```
""")
```

OUTPUT:

Forecasted Prices (Next 5 Days):

	Date	Forecasted Price
0	2018-03-25	1.506894
1	2018-03-26	1.507066
2	2018-03-27	1.507237
3	2018-03-28	1.507409
4	2018-03-29	1.507580

Business Analyst Summary:

- Average avocado prices exhibit clear seasonality and long-term trends.
- Regression analysis shows a steady movement, useful for price planning.
- Prices are expected to increase slightly over the next 6 months.

Recommendations:

- Plan procurement before expected price increases.
- Monitor external risks (weather, logistics, global demand).
- Use more advanced forecasting models for high-accuracy needs (e.g., ARIMA, Prophet).